Debugging Strategies

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Teach debugging through counterexamples and retained evidence.

Provide complete harnesses, distinguish returned values from printed checks and reserve the stock formula for independent testing.

Keep one habit: Look for a discriminating input, a targeted change and an honest coverage limit.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Design revealing tests for a function

Teach Week 12 as a two-session investigation using fictional data. A function can pass one example by coincidence.

Start with a **written rule**, calculate expected results independently, and include ordinary, zero and **boundary** cases. We will test a subtraction function under the contract 0 <= used <= capacity.

Assess the three stated outcomes through the independent task, not by copying the worked result.

By the end, I can test a function with expected values that distinguish a particular bug.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

- Case exposure check: this source example appears in ai-2/6, python-bridge/12. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

- Python 3 editor with a fresh local practice file

- Paper trace table and the student workbook

- Fictional **input** records from this lesson; no real student identifiers

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Loops and Stopping. Give an example and a mistake that the idea helps you avoid.

Look for: Trace variable values before and after each loop iteration

Reference example, not the only acceptable wording: 2, 3 and 4; the stop value is excluded. The body runs three times.

Without opening the old lesson, explain one key idea from Branching Logic. Give an example and a mistake that the idea helps you avoid.

Look for: Trace the selected branch in an if elif else chain

Reference example, not the only acceptable wording: if tests negative first; elif checks <= 12 only after that fails; else handles the rest.

Unindented print runs after any selected branch, whose assignment gives label a value.

7.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Loops and Stopping

Branching Logic

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Intended score is 3 per matching trait. Bug: score starts at 3 and adds 3 for each match. Give expected and actual results for 0,1,2 matches and repair the cause.

Concept to explain

Good **debug**ging isolates a cause with **controlled test**s.

Term	Meaning
bug	A defect that makes a program behave differently from its intended specification.
debug	Find and repair the cause of a program's incorrect behaviour.
controlled test	A test with specified inputs, conditions and expected measurements.
retest	Run a test again after a change or under specified new conditions.

</details>

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach debugging as a fixed-rule comparison with revealing cases

> Provide the full test harness and keep the shelf example distinct from independent stock.

State the subtraction rule and allowed input relationship

A function can pass one example by coincidence. Start with a **written rule**, calculate expected results independently, and include ordinary, zero and **boundary** cases.

We will test a subtraction function under the contract 0 <= used <= capacity.

Read the complete returned-value test harness

```
def remaining(capacity, used):
    return capacity - used

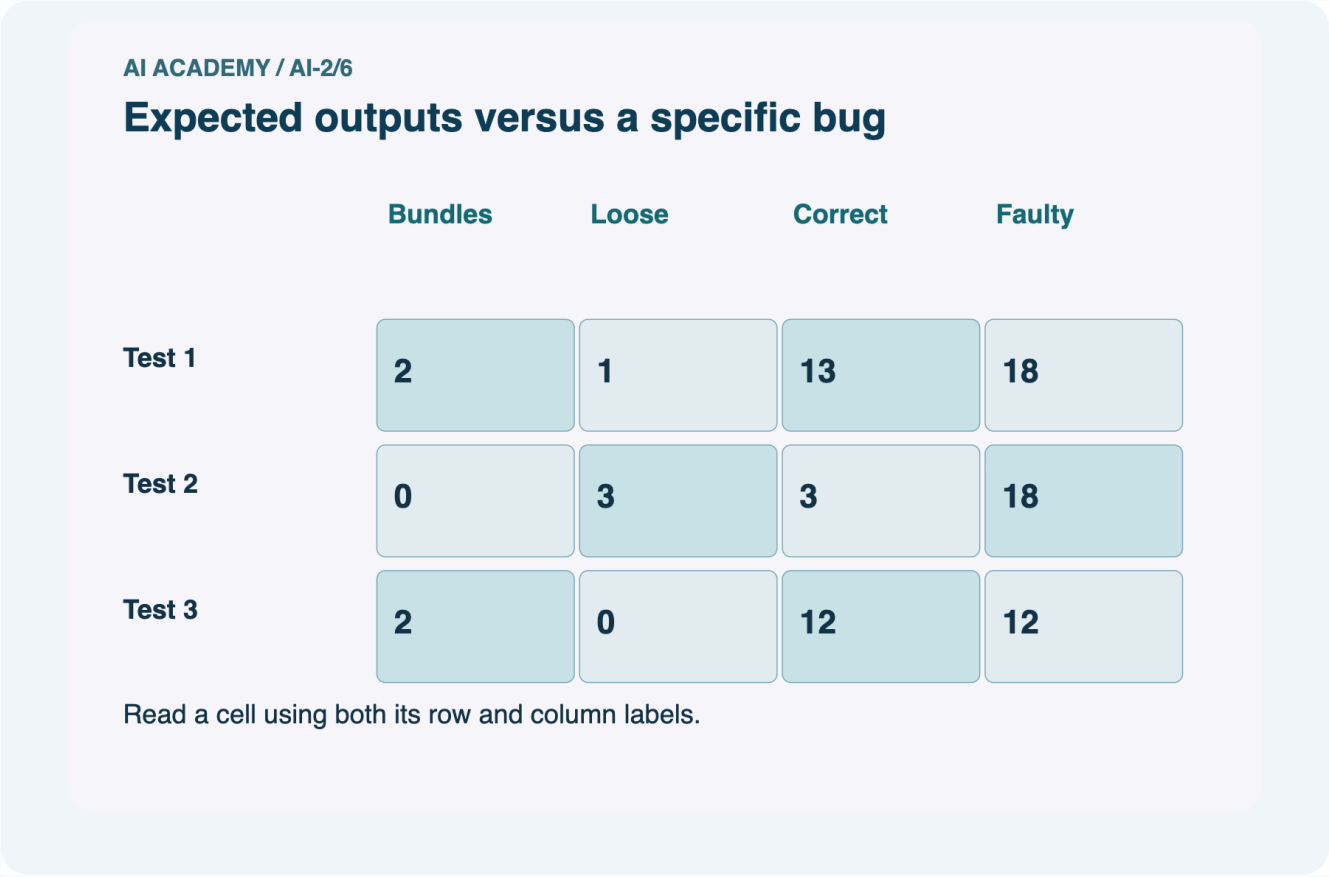
cases = [(8, 3, 5), (8, 8, 0), (8, 0, 8)]
for capacity, used, expected in cases:
```

```
    actual = remaining(capacity, used)
    print(actual == expected)
```

Explain tuples, equality and assertion stopping

Each parenthesized group is a tuple, an ordered group of values. The loop unpacks each group into three names. Equality compares the actual result to the independently calculated expected result.

The worked script prints True three times. An assert actual == expected could instead stop at the first mismatch with **Assertion**Error.



Follow the labelled picture with the matching explanation. Open the vector diagram

Challenge a constant function

Would replacing the body with return 5 pass every worked case? Identify the first revealing test.

Introduce the separate shelf grouping defect

A fictional shelf function should count books in complete bundles plus loose books.

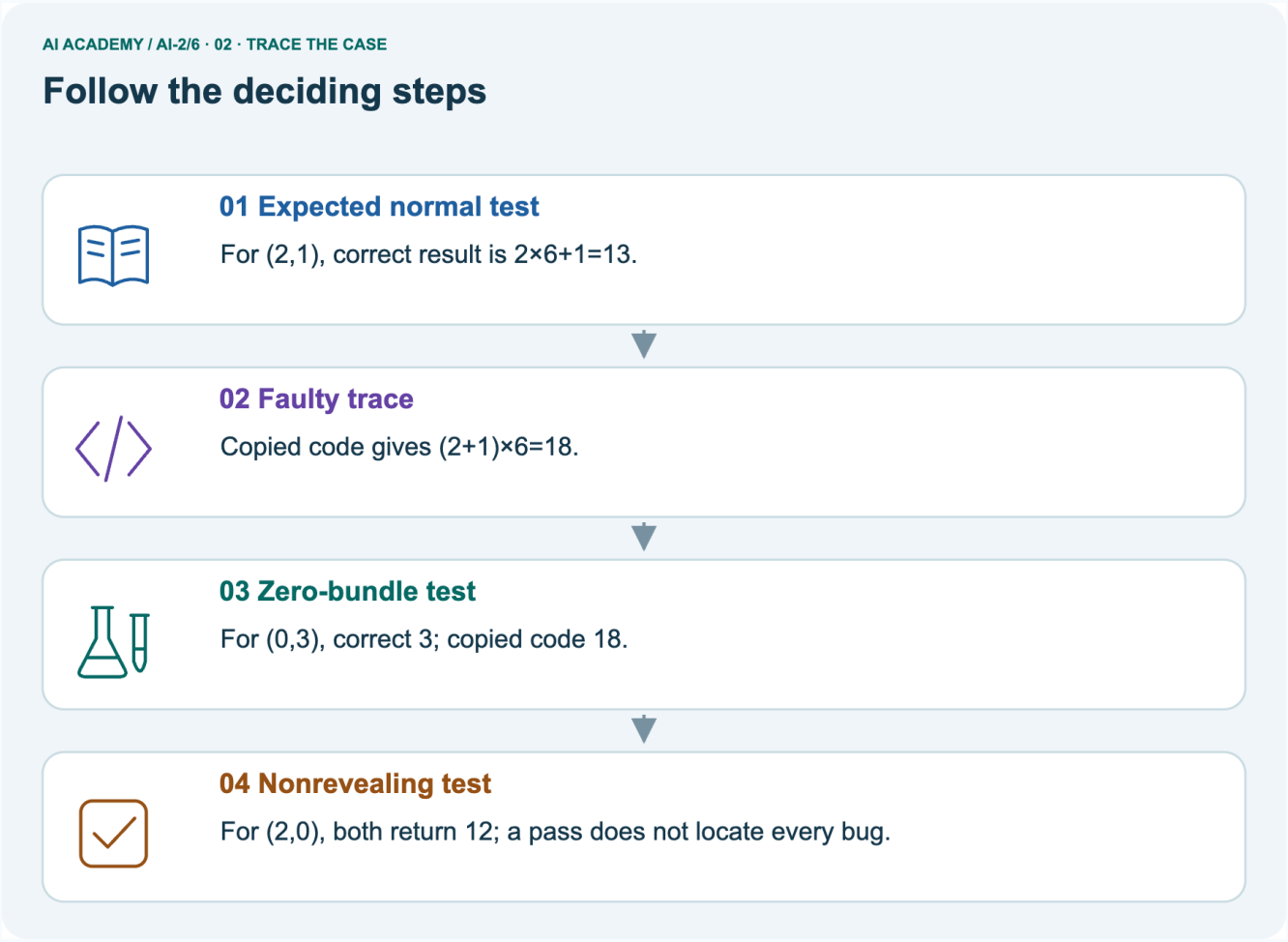
Correct definition: shelf(bundle_count,loose) returns bundle_count*6+loose.

Faulty copied body returns (bundle_count+loose)*6. Predict tests before running.

```
def shelf(bundle_count, loose):
    return (bundle_count + loose) * 6

for bundles, loose in [(2, 1), (0, 3), (2, 0)]:
    print(shelf(bundles, loose))
```

Trace nonzero and zero loose-book cases



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Expected normal test	For (2,1), correct result is 2×6+1=13.
2. Faulty trace	Copied code gives (2+1)×6=18.
3. Zero-bundle test	For (0,3), correct 3; copied code 18.
4. Nonrevealing test	For (2,0), both return 12; a pass does not locate every bug.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Test cases need not all expose the same error. Here loose books must be added after multiplying bundles. A zero-loose case cannot distinguish the two formulas because multiplying zero loose books changes nothing.

Keep expected and actual columns so an observed **output** does not quietly become its own expected answer.

Limit claims about invalid input

These three tests check a specified formula and bug; they do not prove correct handling of every type or invalid request.

Repair only the grouping expression

Change: Repair the body to bundle_count*6+loose.

Prediction: Expected results for (2,1),(0,3),(2,0) are 13,3,12.

Reason: Only bundles are groups of six; loose books are counted individually.

Use pictures to elicit a discriminating test

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Observed: wet-weather case recommends sandals.

2. Divide: **input** correct? yes. rain branch reached? no.

3. Cause: condition says rain = false instead of true.

4. Fix that condition only; rerun the original case and a dry-weather case.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Award two points per matching trait. A bug awards two for every trait.

Test one match and one nonmatch.

Exact explanation to model

Expected 2; buggy result 4. Put the addition inside the match condition. Test zero, one and two matches for 0,2,4.

Explain the added worked example

Key terms

Term	Meaning
Test case	Inputs and an expected result
Boundary	An edge of a permitted range
Regression	A previously working case that fails after a change
Assertion	A check that raises an error when its condition is false

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Debugging Strategies

The facts we will use

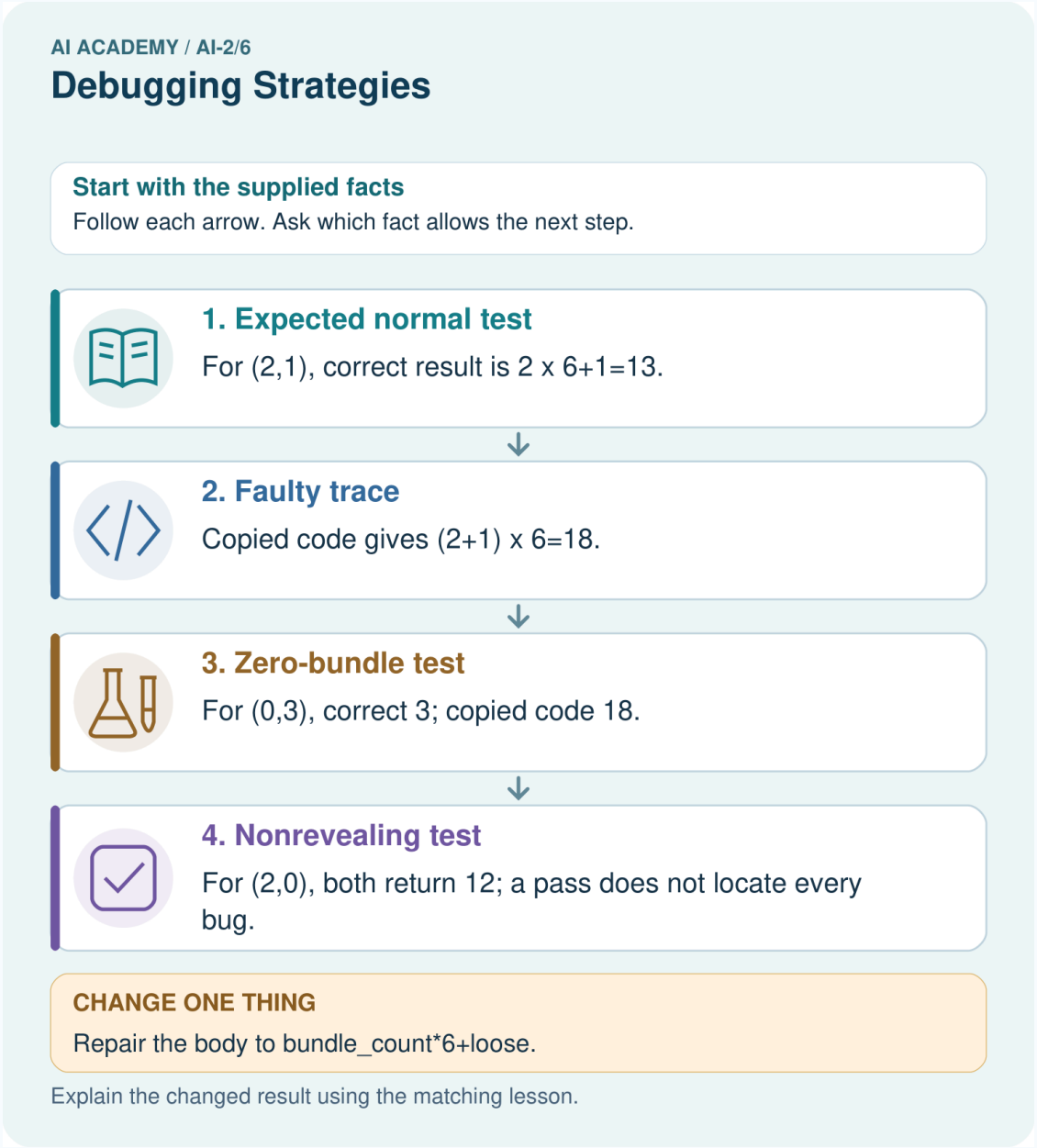
AI ACADEMY / AI-2/6

Expected outputs versus a specific bug

	Bundles	Loose	Correct	Faulty
Test 1	2	1	13	18
Test 2	0	3	3	18
Test 3	2	0	12	12

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: A faulty function always returns 4. Which of your three cases pass and fail?

Expected reasoning and fading help

Teacher check: Only (6, 2) passes. The full and unused cases fail.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for Testing and **debug**ging functions. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Passing this small suite supports these particular cases. It neither proves every allowed **input** correct nor checks invalid **input** handling.

A test outside the contract needs a separately specified expected behavior.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

Which test most directly isolates the handling of loose books, and what should it produce?

(0,3) isolates loose books: expected 3. The faulty body gives 18.

Changed-case reasoning: Expected results for (2,1),(0,3),(2,0) are 13,3,12. Only bundles are groups of six; loose books are counted individually.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: Award two points per matching trait. A bug awards two for every trait.

Write the question to explain: Test one match and one nonmatch.

Reveal after **prediction**s: Expected 2; buggy result 4. Put the addition inside the match condition. Test zero, one and two matches for 0,2,4.

Use the earlier changed question as optional later practice. Reserve Test a new function independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

It passes (8, 3) but fails (8, 8), returning 5 instead of 0. Exit: No. Negative counts are outside our nonnegative-**input** contract and were not tested; rejection behavior must first be defined.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

For remaining's original suite, correct outputs are 5,0,8 with True,True,True; addition gives 11,16,8 with False,False,True; constant 5 gives 5,5,5 with True,False,False. For capacity 6 and used 2,6,0, expectations are 4,0,6; constant 4 passes only the first. The separate faulty shelf gives 18,18,12 instead of 13,3,12; preserve the zero-loose regression case after repair. Actual runs, assertion interruptions and unrun tests must stay explicit. No invalid-input contract is established by these valid cases. Reserve Test a new function independently for the four-per-pack stock formula.

Run and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.
3. Use normal, **boundary**, and unexpected test cards. Mark the first step where expected and actual behavior differ; change one cause; rerun all three.
4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

A successful run proves the conclusion: Passing this small suite supports these particular cases. It neither proves every allowed **input** correct nor checks invalid **input** handling.

A test outside the contract needs a separately specified expected behavior.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: bundles * 3 + singles with (3, 1), (0, 1), (3, 0) expects 10, 1, 9. A constant return 10 passes only the first.

A convenient shortcut preserves meaning: A faulty return capacity + used gives 11, 16 and 8. The unused case passes even though the rule is wrong. Repair the operator to subtraction and rerun all three cases.

Never change an expected result merely to match faulty **output**.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a **boundary** or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Test a new function independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

No. Negative counts are outside our nonnegative-**input** contract and were not tested; rejection behavior must first be defined.

Selected case: Define stock(packs, loose) to return packs * 4 + loose. Create expected tests for (2, 3), (0, 3), (2, 0), then show which detects a faulty packs + loose body.

Judge explanation, trace and limits

Selected case answer

Expected 11, 3, 8. Faulty results 5, 3, 2; first and third reveal it. Use independent expectations in the test loop.

Give one usable repair action

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

W1 Write expectations

Give expected results for remaining(6, 2), remaining(6, 6) and remaining(6, 0).

4, 0 and 6.

Apply the specified subtraction rule to each pair: 6 - 2 = 4, 6 - 6 = 0, and 6 - 0 = 6. Label them ordinary, full and unused cases. These expected answers come from the contract before the program is run.

W2 Find a revealing case

A faulty function always returns 4. Which of your three cases pass and fail?

Only (6, 2) passes. The full and unused cases fail.

Compare the faulty actual result 4 with each independently expected result. It matches the ordinary case expecting 4, but not the full case expecting 0 or unused case expecting 6.

One successful case therefore does not establish that the subtraction rule was implemented.

W3 Repair and retest

Define stock(packs, loose) to return packs * 4 + loose. Create expected tests for (2, 3), (0, 3), (2, 0), then show which detects a

faulty packs + loose body.

Expected 11, 3, 8. Faulty results 5, 3, 2; first and third reveal it. Use independent expectations in the test loop.

W5 Limit the conclusion

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Expected 2; buggy result 4. Put the addition inside the match condition. Test zero, one and two matches for 0,2,4.

Additional transfer practice question

A loop should run four times but runs five. You change its colour and stopping condition at once. Which single change is relevant to test first?

Reasoned practice answer

Test the stopping condition first while preserving the other code. Trace the counter at the **boundary**; changing colour provides no **evidence** about repetition.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Any visible change helps diagnose the cause.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Expected 0,3,6; actual 3,6,9. Initialise score to 0, keeping the per-match addition. The zero-match case isolates the initialisation error. Rerun all three tests; changing only a displayed result would leave the cause intact.

Explanations for the additional workbook examples

W1 Write expectations

Apply the specified subtraction rule to each pair: 6 - 2 = 4, 6 - 6 = 0, and 6 - 0 = 6. Label them ordinary, full and unused cases. These expected answers come from the contract before the program is run.

Only (6, 2) passes. The full and unused cases fail.

A faulty function adds capacity and used. For (6, 2), state actual and expected. Repair it and name a **regression** case.

Actual 8, expected 4. Subtract used; rerun a prior case such as (6, 0), expecting 6.

W4 Test a new function independently

Define stock(packs, loose) to return packs * 4 + loose. Create expected tests for (2, 3), (0, 3), (2, 0), then show which detects a faulty packs + loose body.

All three stock tests pass. Does that prove correct handling of negative counts?

Assess independent transfer

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

When a ranking surprises you, inspect feature values, weights, arithmetic, and sorting separately.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Test the recommender's helper functions with normal, zero and **boundary input**s before **debug**ging the whole application.

Support while preserving the reasoning goal

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, **boundary** or missing field and declare its handling policy before rerunning.

Provide a blank **input**s expected actual table. Extension: use assert for the same cases and explain why only the first failure is reported.

Independent W4 **evidence**: Expected 11, 3, 8. Faulty results 5, 3, 2; first and third reveal it. Use independent expectations in the test loop. Record whether the learner needed no help, a hint or explanation.

Fresh retry: bundles * 3 + singles with (3, 1), (0, 1), (3, 0) expects 10, 1, 9. A constant return 10 passes only the first.

Plan the delayed check

Suggested return points in this level: ai-2/7, ai-2/9, ai-2/12. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 7: Rows, Columns, and Data Types. Retrieve today's **evidence** and identify what carries forward and what is new.

Next week: Rows columns and schemas. Reuse this week's explicit contracts and **evidence**.

▼ **Earlier wording, alternatives and picture transcripts**

Connect and extend learning

Transparent Book Recommender

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Design revealing tests for a function

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

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>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: A faulty function always returns 4. Which of your three cases pass and fail?

Reasoned practice answer: Only (6, 2) passes. The full and unused cases fail.

Check every learner before discussion: Use the worked case for Testing and **debug**ging functions. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Passing this small suite supports these particular cases. It neither proves every allowed **input** correct nor checks invalid **input** handling.

A test outside the contract needs a separately specified expected behavior.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Repair the body to bundle_count*6+loose. Require a **prediction** and a reason before comparing. Expected result: Expected results for (2,1),(0,3),(2,0) are 13,3,12.

Reason: Only bundles are groups of six; loose books are counted individually.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-**boundary** and indexing lessons deliberately include failure cases.
</details>

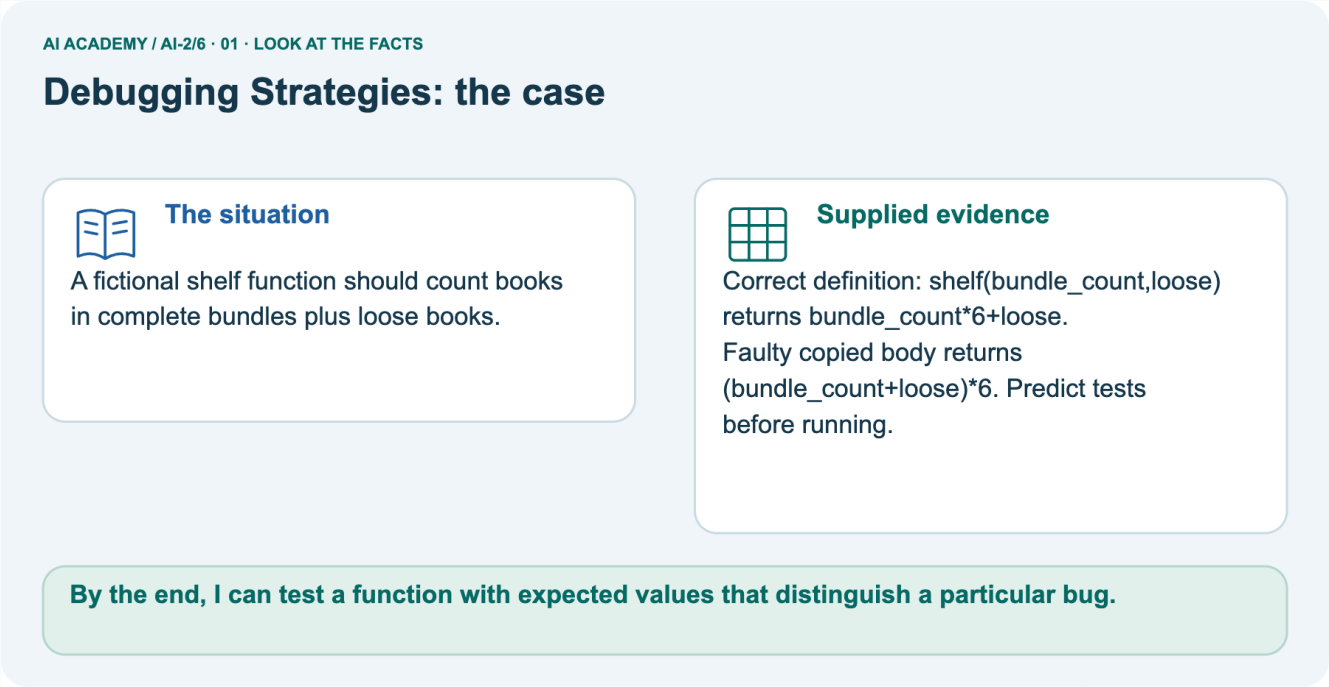
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Debugging Strategies: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

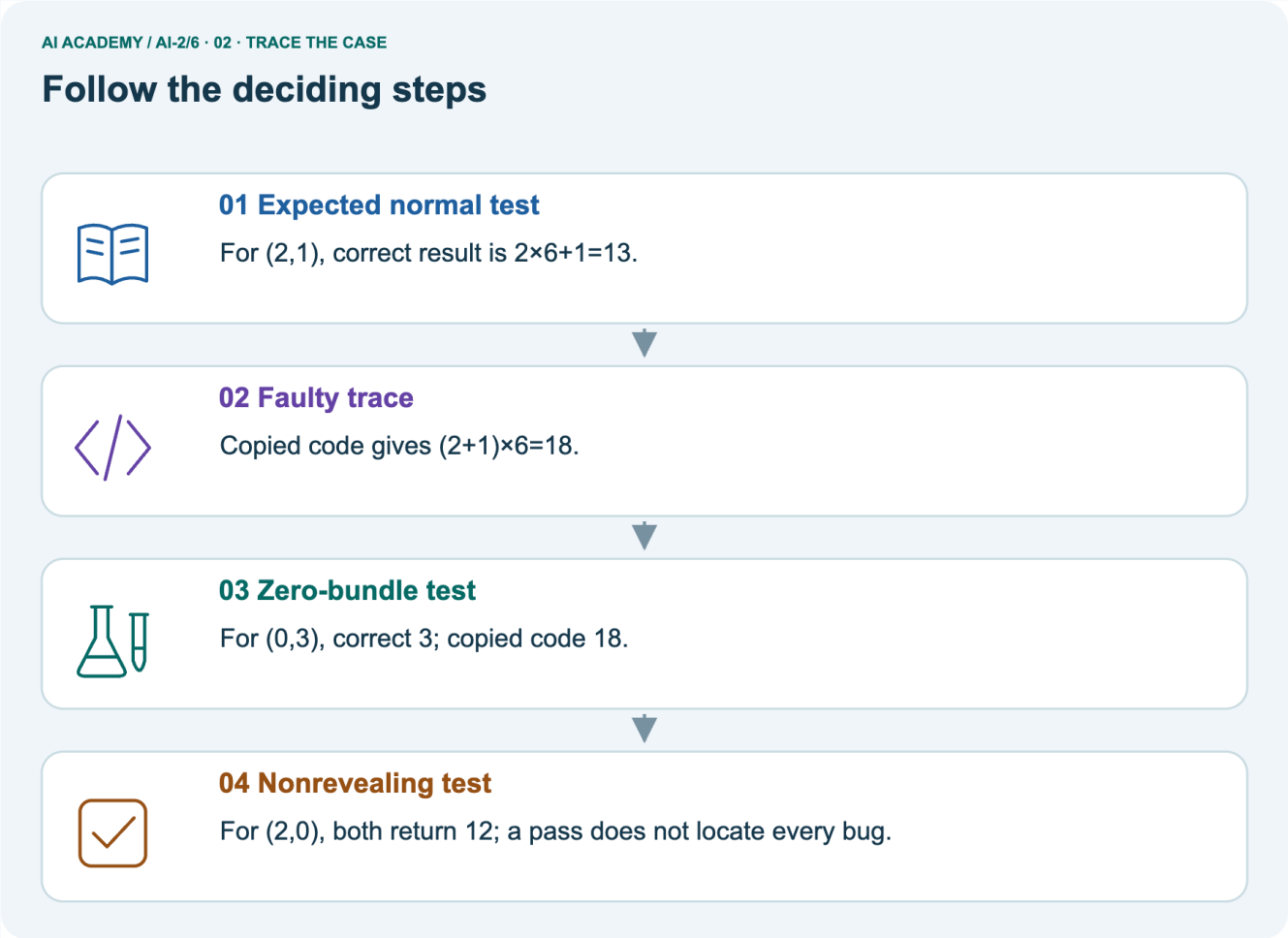
A fictional shelf function should count books in complete bundles plus loose books. Supplied facts: Correct definition: shelf(bundle_count,loose) returns bundle_count*6+loose. Faulty copied body returns (bundle_count+loose)*6.

Predict tests before running.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



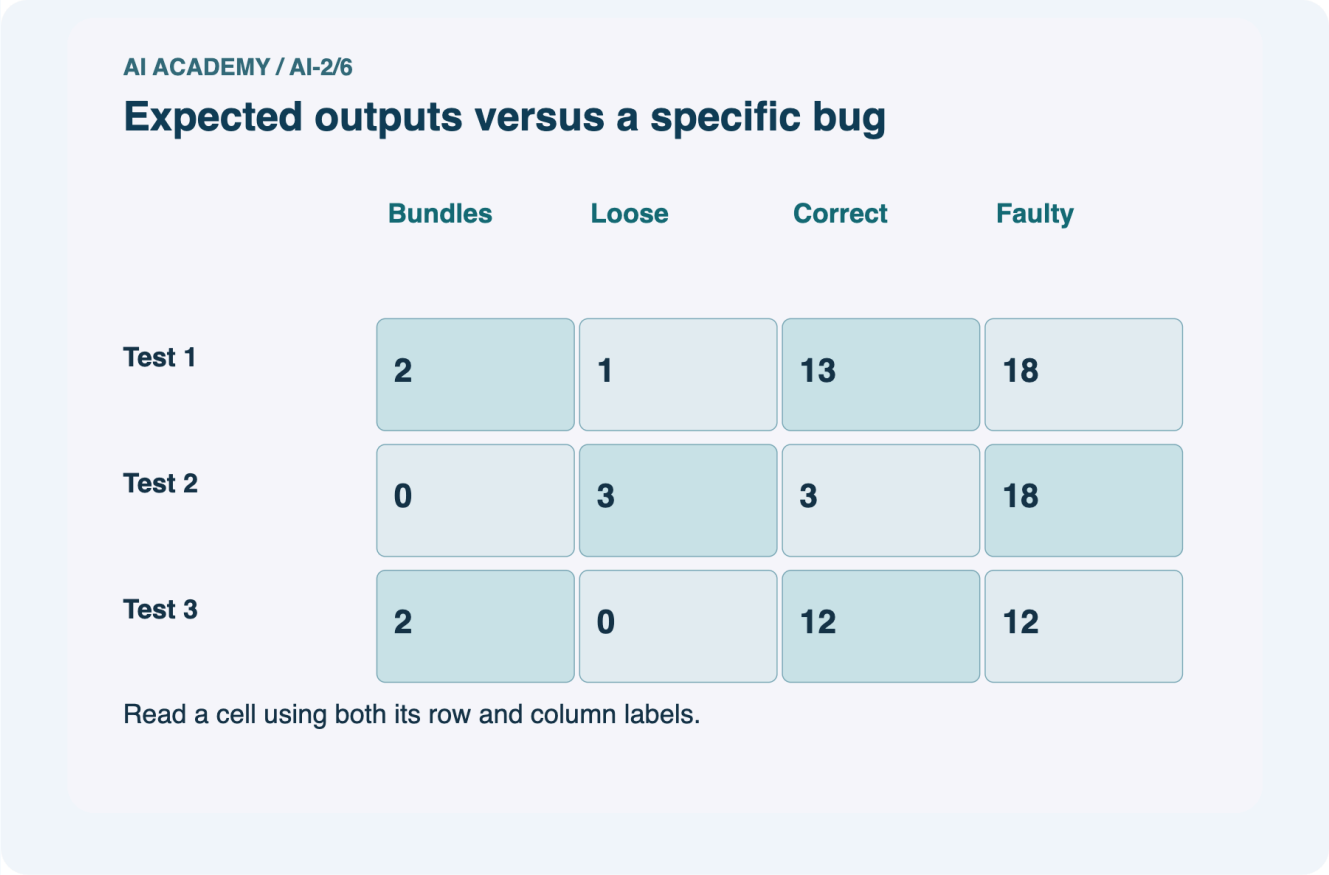
Follow the labelled picture with the matching explanation. Open the vector diagram

Expected normal test: For (2,1), correct result is 2×6+1=13. → Faulty trace: Copied code gives (2+1)×6=18. → Zero-bundle test: For (0,3), correct 3; copied code 18. → Nonrevealing test: For (2,0), both return 12; a pass does not locate every bug.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Expected outputs versus a specific bug



Follow the labelled picture with the matching explanation. Open the vector diagram

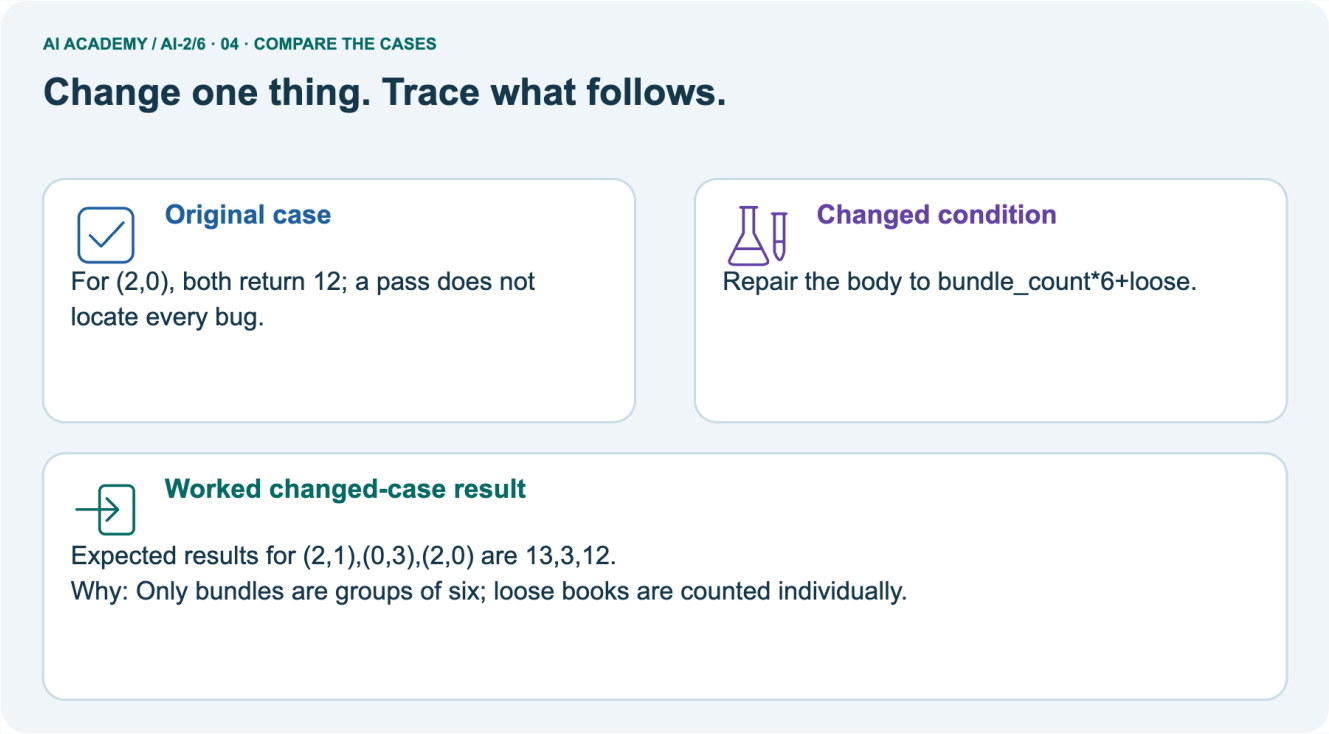
Expected **output**s versus a specific bug. Test 1: Bundles = 2, Loose = 1, Correct = 13, Faulty = 18. Test 2: Bundles = 0, Loose = 3, Correct = 3, Faulty = 18. Test 3: Bundles = 2, Loose = 0, Correct = 12, Faulty = 12.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: For (2,0), both return 12; a pass does not locate every bug. Changed condition: Repair the body to bundle_count*6+loose. Expected result: Expected results for (2,1),(0,3),(2,0) are 13,3,12.

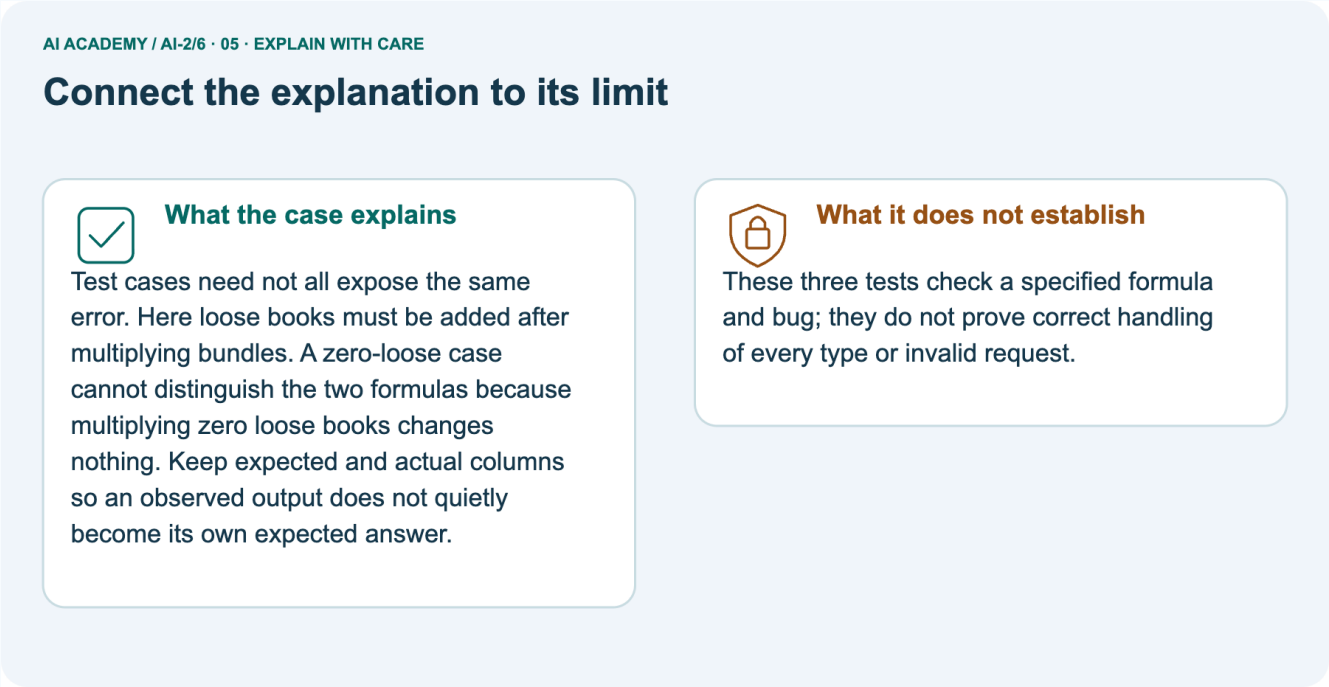
Reason: Only bundles are groups of six; loose books are counted individually.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Only bundles are groups of six; loose books are counted individually.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Test cases need not all expose the same error. Here loose books must be added after multiplying bundles. A zero-loose case cannot distinguish the two formulas because multiplying zero loose books changes nothing.

Keep expected and actual columns so an observed **output** does not quietly become its own expected answer.

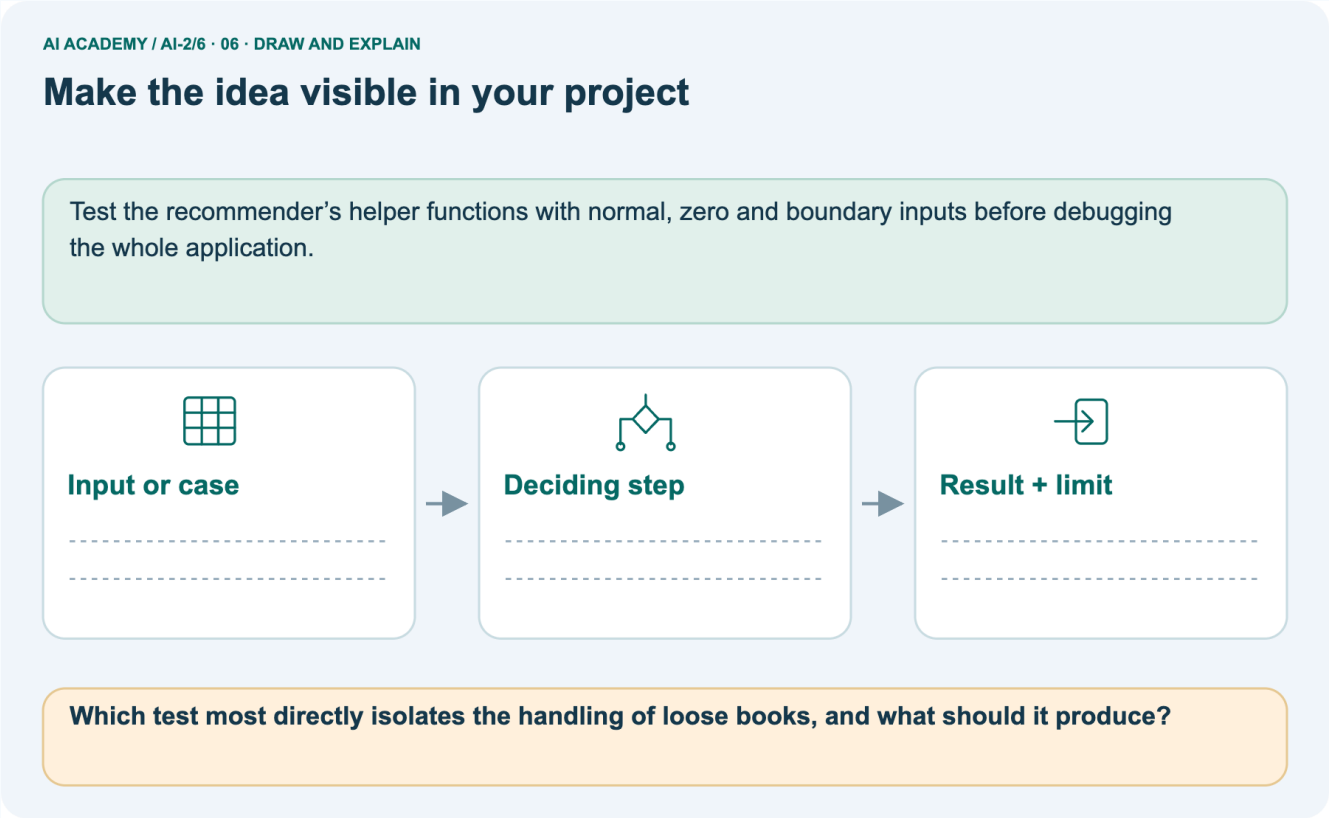
Limit: These three tests check a specified formula and bug; they do not prove correct handling of every type or invalid request.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: These three tests check a specified formula and bug; they do not prove correct handling of every type or invalid request.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Test the recommender's helper functions with normal, zero and **boundary input**s before **debug**ging the whole application. Draw your **input**, deciding step, result and limit.

Supported question: Which test most directly isolates the handling of loose books, and what should it produce?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: (0,3) isolates loose books: expected 3. The faulty body gives 18.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Debugging Strategies: the case

02. Follow the deciding steps

03. Expected outputs versus a specific bug

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.